

Brian Yin

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EDUCATION

University of Waterloo

Bachelor of Computer Science (BCS)

- GPA: 91.00%

Waterloo, ON

Sep. 2025 – May 2030

EXPERIENCE

University of Waterloo (Cynorix Inc.)

Networking Research Intern

May 2026 – Present

Waterloo, ON

- Analyzed one-way latency travel times of **4M+** TCP/UDP packets across **40+** hours of testing to identify periodic Starlink handoff behavior, pinpointing handoff timing to precise UTC offsets and spacing intervals
- Prototyped a predictive dual-path switching system in **Python** that preemptively routes TCP packet traffic over cellular during Starlink handoffs, reducing handoff-induced latency spikes by **~95%**
- Synchronized Linux and **AWS EC2** clocks via **chrony**, achieving **sub-millisecond** timing accuracy for one-way latency measurements across Starlink and cellular paths
- Developing a secure encryption method exploiting **UDP** packet error events over cellular networks for key generation

UWaterloo Orbital

Full-Stack Software Developer

Sep. 2025 – Present

Waterloo, ON

- Architected a singleton **Pydantic** configuration manager for **UW Orbital's CubeSat** ground station, replacing **14+** hardcoded backend configuration settings with a centralized, type-safe validation system
- Implemented **5 FastAPI endpoints** for ARO picture requests using **SQLModel** ORM and **Pydantic**
- Wrote **30+ unit tests** using **Pytest** to validate environment variable overrides and configuration integrity across development environments, leading to **100% code coverage** for the configuration manager

PROJECTS

TCP Layer 4 Load Balancer 🚀 | *C++, Golang, React, TypeScript, Linux (WSL)*

- Architected a **C++** Layer 4 TCP load balancer using **epoll**-based event handling, **multithreading**, and mutexes, with round-robin backend routing validated against a dockerized nginx backend fleet
- Built a load-testing agent in **Go**, using goroutine-based concurrency, with token-bucket rate limiting, and bounded worker pools to compute percentile latency metrics under load
- Implementing a **React.js** and **Typescript** dashboard streaming latency results using **Websockets** and generating graphs using Recharts.js

Self-Driving RC Car 🚗 | *Python, OpenCV, Raspberry Pi*

- Led a team of **3** to develop an autonomous Raspberry Pi 4 RC car, placing **1st nationally** and **5th internationally** while representing **Team Canada** at the 2024 WRO Future Engineers Final
- Developed a multistage **OpenCV** vision pipeline utilizing LAB thresholding, image blurring and morphological filtering to refine and extract obstacle contours at **30 FPS**
- Designed PD steering controller with dynamic ROI optimization for reliable autonomous wall-following, obstacle avoidance, and parking, resulting in a **95% test run success rate**

Punch Perfect 🥊 | *React, Node.js, Express.js, Supabase (PostgreSQL), MediaPipe.js*

- Deployed a webcam-based real-time boxing game using **MediaPipe** pose estimation to process body landmarks at **30 FPS** with **sub 150ms** detection latency on consumer hardware
- Engineered a punch-detection computer vision pipeline to classify punches from pose data with **85% accuracy**
- Integrated a **RESTful API** with **Express.js** and **Supabase Auth**, implementing JWT-secured endpoints and **rate limiting** to ensure leaderboard and score integrity
- Architected a modular **React** frontend utilizing custom hooks, enabling **80% code reuse** across **four** game modes

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript, Golang, SQL, HTML/CSS, Bash

Frameworks / Libraries: React, Tailwind, Node.js, Express.js, FastAPI, OpenCV, MediaPipe, Pytest

Tools / Development: Git, GitHub, VS Code, Linux, Arduino IDE, Raspberry Pi, Supabase, Vercel, Render, Vite

Databases: MySQL, PostgreSQL, SQLite